Malitha Gunawardhana

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Summary

Machine Learning Engineer, AI Engineer and Data Scientist with 5 years of professional experience across machine learning, AI and software engineering, complemented by a PhD in deep learning, computer vision and model reliability. Built end-to-end systems across generative AI, agentic AI, computer vision, time-series and sensor analytics, spanning data preparation, experimentation, APIs, deployment, monitoring and failure analysis. Delivered measurable outcomes including more than 70% reduction in seismic processing effort, more than 5% reduction in false alarms across 3,000+ cameras and 90%+ success in a low-latency production platform. Strong in Python, software engineering, MLOps, cloud deployment, statistical evaluation and stakeholder collaboration. Seeking full-time roles across AI engineering, machine learning, data science and software engineering where I can build reliable, production-focused systems that solve complex real-world problems.

Professional Experience

Machine Learning Engineer (Part-Time) Mar 2026 - Sep 2026
Viotel Pty Ltd *Parnell, Auckland*

- Developed and deployed an end-to-end deep-learning system for automated seismic phase identification across **3,700+ events and 30,000+ station samples**, reducing geophysicists' manual processing time by **more than 70%**.
- Analysed high-frequency sensor data across **10+ telecommunications towers**, combining environmental context, controlled experiments and threshold tuning to separate abnormal vibration from normal variation and progress a proof of concept to **alpha testing on four towers**.
- Packaged inference with **Docker**, automated recurring processing on AWS-hosted data, and performed sensor- and time-stratified failure analysis to distinguish model errors from hardware faults, timing issues and site noise.

Machine Learning Engineer (remote from Dec 2023) Sep 2022 - Jan 2025
IPPT-PAN *Warsaw, Poland*

- Developed deep-learning models for breast-tumour segmentation, benign/malignant classification and BI-RADS prediction using a curated multi-centre dataset of **3,000+ ultrasound images**, achieving **more than 86% Dice** for tumour segmentation.
- Built reusable data and ML workflows for sourcing, curation, preprocessing, label/data-quality checks, training and evaluation, converting research notebooks into reproducible **AWS and Docker-based pipelines**.
- Evaluated performance, bias and generalisation across hospitals, datasets, scanner differences and patient subgroups using cross-validation, cross-dataset testing and systematic error analysis.

Artificial Intelligence Researcher Sep 2022 - Sep 2023
Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) *Abu Dhabi, UAE*

- Conducted research across self-supervised video learning, semi-supervised learning and image-model calibration, designing controlled experiments and benchmarking methods for generalisation and predictive reliability.
- Co-developed a train-time object-detection calibration method that reduced calibration error against baselines on in-domain and out-of-domain benchmarks, resulting in a **CVPR 2023 publication**.

Machine Learning Engineer Jun 2022 - Nov 2022
PromiseQ GmbH *Berlin, Germany*

- Developed and optimised real-time computer-vision models for an AI-surveillance platform operating across **3,000+ CCTV cameras**, covering detection, tracking and safety/security use cases.
- Reduced false alarms by **more than 5%** through model calibration, error analysis, retraining and improvements to detection, tracking and post-processing logic.
- Monitored deployed-model performance, input quality, drift, confidence, latency, incidents and model versions; debugged production ML workflows across **GCP and AWS**, streaming data, APIs and GitHub-based CI/CD.

Full-Stack Software Engineer Mar 2021 - May 2022
Xeptagon (Pvt) Ltd *Colombo, Sri Lanka*

- Engineered a low-latency production platform using **Go, MongoDB, REST APIs, TypeScript and JavaScript**, achieving a **90%+ successful domain catch rate** with successful catches completed within approximately **50 ms** of availability.
- Translated ambiguous requirements from CEOs, CTOs and managers into backend services, APIs, database schemas, logging and production tooling while diagnosing live-system behaviour as requirements evolved.
- Directed a **four-member team** across architecture decisions, project planning, technical documentation and code review.

Software Engineer (Intern) Jun 2019 - Dec 2019

Synergen Technology Labs (Pvt) Ltd *Colombo, Sri Lanka*

- Developed signal-processing and machine-learning pipelines for four-class physiological stress classification, achieving **more than 90% accuracy** on a held-out participant test set in an internal proof of concept.

Education

University of Auckland Auckland, New Zealand

Doctor of Philosophy (Deep Learning, Computer Vision & Data Science) *Dec 2023 - Aug 2026*

- Thesis: Robust, Uncertainty-Aware, and Scalable Deep Learning for Bi-Atrial Segmentation from LGE-MRI in Atrial Fibrillation. Award: Health Research Council Scholarship.
- Built the full ML lifecycle across multi-centre clinical datasets, including sourcing and curation, annotation and label-quality assurance, preprocessing, training, experiment tracking, systematic evaluation, reproducibility and failure analysis.
- Evaluated robustness, generalisation, uncertainty and model risk under distribution shift using controlled experiments, confidence intervals, statistical testing and clinically meaningful error analysis.
- Developed self-supervised representation-learning workflows and reproducible research infrastructure using HPC, multi-GPU training, Docker, automated jobs and version control, collaborating with clinicians and progressing the work to a clinical research prototype.

University of Moratuwa Moratuwa, Sri Lanka

B.Sc. Engineering Honours *Jan 2017 - Jul 2021*

Selected Technical Projects

Healthcare Voice and Chat Assistant Personal Project

Agentic AI, RAG, APIs, observability and cloud deployment

- Developed an end-to-end healthcare assistant with a **FastAPI** backend and React/TypeScript interface, designing **LangGraph stateful multi-agent workflows** with guardrails, routing, structured outputs, tool calling and **MCP** integrations.
- Integrated Qdrant/pgvector RAG, memory and Redis/Arq background processing, instrumented route/tool/latency behaviour with **Langfuse**, and deployed with Docker and **GitHub Actions CI/CD to AWS ECS/Fargate**.

ChurnOps: Customer Churn Prediction and MLOps Personal Project

Predictive modelling, data engineering and production ML

- Built reusable pandas/scikit-learn and PySpark/MLlib pipelines connecting validation, preprocessing, model selection and evaluation to batch and Kafka-based streaming inference, with **MLflow** experiment tracking and model registration.
- Implemented data-quality and drift monitoring, Kafka micro-batch inference, Airflow DAGs and Dockerised AWS deployment workflows using **S3, ECR, ECS/Fargate, RDS and CloudFormation**.

Technical Skills

Programming Languages: Python, SQL, Go, TypeScript, JavaScript, Bash, MATLAB, R
Machine Learning: PyTorch, TensorFlow, scikit-learn, Computer Vision, Medical Imaging, Time-Series Modelling, Anomaly Detection, Self-/Semi-Supervised Learning, Model Calibration
Generative AI and LLMs: Hugging Face Transformers, RAG, LangChain, LangGraph, MCP, Tool Calling, Structured Outputs, LLM Evaluation, LoRA/QLoRA, GraphRAG, Qdrant, pgvector, Langfuse
Data and Distributed Computing: PySpark/MLlib, Kafka, Airflow, PostgreSQL/RDS, Redis, MongoDB
Data Analytics and BI: Power BI, Excel, Statistical Analysis, Hypothesis Testing, Data Visualisation, Data Quality Assurance
Developer and AI Tools: Docker, AWS, GCP, Azure, MLflow, FastAPI, REST APIs, GitHub Actions, Git, Linux, Jira, Confluence
Technical Domains: Deep Learning, Generative AI, Agentic AI, Computer Vision, MLOps, Model Deployment and Monitoring, Signal Processing, Predictive Modelling, Robustness and Generalisation

Service and Leadership

Auckland Bioengineering Institute, University of Auckland 2024 – 2026

- Elected Student Council Member representing postgraduate students in institute discussions and student initiatives.

Sustainable Education Foundation 2024 – 2025

- Mentored students in academic, professional and personal development.

Rotaract Club, University of Moratuwa Alumni 2016 – 2024

- Vice President, Club Service and former Club Service Director, leading member engagement, fellowship and community initiatives; recipient of multiple Spirit of Service awards.